

Will water usage from AI decrease overtime as AI develops?

Posted by u/CertainPass105 • 0 points • 24 comments

Currently AI uses an insane amount of water (which is one of our most scarce resources) Do you think water consumption will get more efficient regarding AI quickly or will this just add to environmental degradation.

Comments

u/behindmyscreen_again • 1 points • 2025-02-07 20:29 UTC

Considering the way Deepseek works is to modularize expert knowledge and reduce "accuracy" to be "good enough" which resulted in a significant processing and power drop, I think it's safe to say that's a direct things are going to go.

u/Sirisian • 1 points • 2025-02-07 20:03 UTC

Rule 2, please post AI questions during the weekends.

u/[deleted] • 1 points • 2025-02-07 20:02 UTC

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u/[deleted] • 1 points • 2025-02-07 20:01 UTC

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u/Gelkor • 3 points • 2025-02-07 19:57 UTC

Everybody here missing the point that water used by data centers is coming from water utility providers, which generally come from aquifers that would otherwise be used in agriculture or as drinking water by the local community.

Yeah, the water doesn't "disappear", but it is being drained from the aquifers faster than it would nominally be used which can lead to droughts and negative impacts on agriculture and available drinking water for local communities. Also see: Arizona's huge problem with groundwater being used for growing alfalfa that is then shipped overseas.

If data centers were pulling from a river it would be one thing, but they are generally plumbed into local utilities.

u/FaultElectrical4075 • 2 points • 2025-02-07 19:53 UTC

The average 18-hole golf course uses literally 400 times more water per day as the average datacenter, if I'm being generous(5 million gallons for a datacenter vs 2 billion for a golf course)

Also, datacenter water is recycled. It doesn't just disappear.

But yes, it will decrease over time as AI algorithms/hardware become more optimized

u/KidKilobyte • 0 points • 2025-02-07 19:50 UTC

Here's an idea, put the data cloud in orbit above the clouds and use solar energy there. Launch costs are set to plummet with SpaceX and plans are in the works already to develop solar energy from orbit, rather than deploy solar in orbit and beam microwaves to Earth (which may have environmental impacts), consume the energy in place more efficiently. Granted it will need huge radiators, but they'd be needed if you were beaming the power back to Earth anyway.

u/zandadoun • 4 points • 2025-02-07 19:50 UTC

Water doesn't disappear. Earth has had the exact same amount of water for millions of years.

u/mjzimmer88 • 1 points • 2025-02-07 20:02 UTC

It's had ALMOST the exact same amount of water for millions of years.
On the scale of water on this planet its less than a rounding error though.

For example:

Some 500 gallons or so are up on the ISS.

u/Canna_Milf • 1 points • 2025-02-07 19:47 UTC

Start building data centers at the bottom of the ocean

u/euph_22 • 1 points • 2025-02-07 19:53 UTC

Microsoft is way ahead of you: [https://en.wikipedia.org/wiki/Project_Natick]
(https://en.wikipedia.org/wiki/Project_Natick)

u/LEADEROFTHEPEOPLE • -1 points • 2025-02-07 19:46 UTC

See the world for what it should be, join r/LEADINGTHEPEOPLE

u/Thoguth • 2 points • 2025-02-07 19:45 UTC

How is it using water? You mean for electricity or cooling, to run? Or like for manufacturing, as part of the industrial processes that create chips?

u/The_mingthing • 3 points • 2025-02-07 19:45 UTC

This is so ridiculous! The water does NOT disappear!!!!!!!

Water used for cooling needs to have biocides added to it to keep it from clogging up with bacteria/Alge, and corrosion inhibitors to keep the piping from being corroded! You dont dump it out, you run it trough radiators to cool it back down again and run it back trough the system. Like IN A CAR!

u/ivanhoe90 • 7 points • 2025-02-07 19:43 UTC

Are you talking about hydroelectric power plants, which generate the electricity (part of which could be used by the AI)?

This water does not disappear, it just falls from a high point to a low point. At some point, it will be warmed by the sun, it will evaporate, and the rain will fill the river again.

This energy does not come from the water, it comes from the sun, which does the "hard job" of moving the water up from the sea to the atmosphere.

u/rileystoneill • 6 points • 2025-02-07 19:43 UTC

The water is not consumed. It is part of the watercycle, if the heat from AI is cooled by water then the water turns to vapor where it returns in the form of rain.

u/Aggressive_Salad_293 • 15 points • 2025-02-07 19:40 UTC

Are they shooting the water into outerspace afterward?

u/KidKilobyte • -2 points • 2025-02-07 19:43 UTC

Excellent point, if data centers are built upstream of where it is needed, should be minimal impact other than some minor evaporation (assuming not using ground water).

u/behindmyscreen_again • 1 points • 2025-02-07 20:30 UTC

Hot water comes out. It needs to be cooled before it's returned to the environment.

u/Aggressive_Salad_293 • 3 points • 2025-02-07 19:46 UTC

Do you know what happens to evaporated water? Water on earth is a closed system. It may need some treatment but unless it's being converted into other compounds I'm not sure what we're even discussing.

u/kooshippuff • 1 points • 2025-02-07 19:54 UTC

There was a bit of fearmongering around AI that said each prompt given to ChatGPT consumed the equivalent of a bottle of water or something like that, which then has people concerned it's going to exacerbate a water shortage.

But like, the water is a thermal mass used too move heat around. It literally just warms up and cools down. A personal computer can use the same liter of water in a closed loop for years.

I don't know how datacenters do it, but if they have some means of cooling the water back down, like big radiators or something, it's probably the same deal- they just reuse the same water indefinitely. If they were to dump the hot water in the sewer or something, that'd put added stress on water treatment plants, but to mention being a bit of a dick move, but still- it's not like the water is going anywhere.

u/[deleted] • 1 points • 2025-02-07 19:40 UTC

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u/Spiritual_Carob_7512 • 1 points • 2025-02-07 19:39 UTC

Water usage will go up, then it will level out. It won't go down.

u/duckdodgers4 • 3 points • 2025-02-07 19:42 UTC

I have a bad feeling about this